

SOFT COMPUTING CAPSTONE

SC02

Smart Washing Machine Decision and Resource Optimizer

Beginner Project Guide

All steps at a glance, with a fully spoon-fed Step 1.

Team size: 3-4 students

Step 1 target: 3-5 working days

M1: one week before mid-semester examination

M2: one week before end-semester examination

What are you building?

Build an app that recommends wash time, water, detergent, and spin speed after the user describes the clothes and water availability.

Who will use it?

A washing-machine user who needs safe settings without wasting water or detergent.

One-sentence problem

Given dirt, load, fabric, and water availability, recommend wash time, water, detergent, and spin speed.

What M1 must show

A fuzzy washing controller, compared with fixed washing presets.

What M2 must add

Use GA to balance washing quality with water, detergent, and energy use, then deploy the app.

Important

M1 must be a working Product V1: input -> processing -> output. Step 1 is the preparation needed before the main fuzzy, ANN, or GA algorithm. M2 must add a real technical improvement and a deployed Product V2.

Your eight-step project path

Step	Stage	What happens
1	M1	Choose the washing machine capacity and define inputs.
2	M1	Create and validate realistic washing scenarios.
3	M1	Build fixed wash-cycle presets.
4	M1	Build fuzzy variables, rules, and outputs.
5	M1	Join the controller and screen as Product V1.
6	M2	Use GA to tune quality and resource trade-offs.
7	M2	Test delicate, scarce-water, and heavy-load cases.
8	M2	Deploy Product V2 and finish all evidence.

Where this guide spends most of its time

The remaining pages explain Step 1 in very small, doable actions. Finish every checkbox before starting the main algorithm.

Do not begin fuzzy rules, neural-network training, or GA optimization just because one member is excited to code. First agree on the problem, fields, units, sample data, baseline, tests, and evidence.

What Step 1 must produce

- ☐ A GitHub repository that every member can open.
- ☐ A README.md that explains the project and how the team will work.
- ☐ A docs/STEP-1.md project contract.
- ☐ A data/README.md with field meanings, units, sources, and assumptions.
- ☐ The correct starter data files.
- ☐ Twenty clean starter rows or one small, complete fixture.
- ☐ Five named test cases, including an invalid or impossible case.
- ☐ A simple baseline written as numbered rules or pseudocode.
- ☐ A validation program that prints PASS for valid data and rejects bad data.
- ☐ A simple Product V1 screen sketch.
- ☐ At least one meaningful Git commit from every member.
- ☐ A three-minute faculty demonstration.

Suggested five-day plan

1	Day 1: create repository, add members, assign roles, and write the one-sentence problem.
2	Day 2: define every input/output field, unit, range, category, and missing-value rule.
3	Day 3: create the 20-row sample or small fixture; write the five test cases.
4	Day 4: write baseline rules, validation code, screen sketch, and make member commits.
5	Day 5: clone/run from the beginning, correct errors, and rehearse the faculty demo.

Create the working space

1. One member creates a new repository. Suggested name:

SC02-Smart-Washing-Machine-Decision-and-Resource-Optimizer-Team-Name

2. Add all members as collaborators. Every member accepts the invitation.

3. Add these folders and files:

```
README.md
requirements.txt
docs/STEP-1.md
data/README.md
data/sample_input.csv # or the project-specific starter files
src/validate_data.py
notebooks/
app/
results/
report/
```

4. Assign first responsibilities

Responsibility	Exact Step 1 job
Repository	Create folders, add members, and test clone/run steps.
Data	Write fields/units, make the sample, and record sources.
Baseline	Write simple rules and expected results for five cases.
Testing/UI	Write validation, draw the screen, and collect evidence.

For three students, combine Repository with Testing/UI. These are first responsibilities, not permanent silos. Everyone must understand the complete Step 1.

5. If using Git commands

```
git clone REPOSITORY-URL
cd REPOSITORY-NAME
git checkout -b your-name-step1
# create or edit your assigned files
git add .
git commit -m "Describe the real work you completed"
git push -u origin your-name-step1
```

If Git commands feel difficult, use GitHub Desktop. The required idea is the same: pull first, work on a small task, write a clear commit message, push, and let the team review.

Write docs/STEP-1.md

Copy these headings into **docs/STEP-1.md**. Then replace every instruction with your team's answer.

Step 1 Project Contract

Team and responsibilities

- Member name - first responsibility

One-sentence problem

Given dirt, load, fabric, and water availability, recommend wash time, water, detergent, and spin speed.

User of the product

A washing-machine user who needs safe settings without wasting water or detergent.

Inputs and units

Write the field table from the next page.

Outputs and units

wash_time_min, water_litre, detergent_ml, spin_rpm

Baseline method

Write the numbered baseline rules from this guide.

Soft Computing method for M1

A fuzzy washing controller, compared with fixed washing presets.

Advanced method for M2

Use GA to balance washing quality with water, detergent, and energy use, then deploy the app.

Dataset/scenario sources

Add real URLs, licences, and assumptions.

Five mandatory test cases

Copy and complete the five cases in this guide.

Product V1 screen sketch

Insert a photo or exported image of the sketch.

Risks and assumptions

List what may be wrong, missing, simulated, or temporary.

Step 1 completion evidence

Add links to files, commits, and validation output.

Before saving

[] Replace generic words such as 'user' or 'data' with project-specific words.

[] Write units beside every number.

[] Mark temporary thresholds and assumptions as temporary.

[] Do not claim that simulated data is real data.

[] Do not paste a source URL without saying what information came from it.

Create the data dictionary

Field	Simple meaning	Type/unit	Starter rule
scenario_id	Unique wash-case name	text	SC02-001, SC02-002, ...
dirt_score	How dirty the clothes are	number	0 to 10
load_kg	Weight of clothes	kg	0.5 to chosen machine capacity
fabric_type	Main fabric	category	delicate, cotton, synthetic, heavy
water_availability_pct	Water available	percentage	0 to 100

For every field, the team must answer

- ☐ What does this field mean in one simple sentence?
- ☐ Is it a number, category, text identifier, input, target, or output?
- ☐ What is its unit?
- ☐ What values are allowed for the 20-row sample?
- ☐ What should happen when the value is missing?
- ☐ Which source or assumption justifies the final range?

Write the outputs separately

Required outputs: wash_time_min, water_litre, detergent_ml, spin_rpm

Source and 10,000-record plan

Choose one real machine capacity. Cite safe ranges or product documentation. Later generate at least 10,000 valid wash cases.

Step 1 uses only a small sample. The final 10,000+ records/scenarios are created in Step 2, after the small sample and validation are correct.

Create the first sample carefully

Do not begin with 10,000 rows. First make a small file that every member understands.

How to create the sample

1	Create the file or files listed on the repository page.
2	Type the field names exactly as written in the data dictionary.
3	Create five rows for the mandatory cases shown below.
4	Add five ordinary rows with different valid values.
5	Add five boundary rows close to minimum or maximum values.
6	Add five difficult rows that combine conditions.
7	Keep one wrong row in a separate test file. Do not mix it into valid data.
8	Save as CSV using comma-separated values and a header row.

Five mandatory starter cases

Name	Example input	What to check
SC02-001	2, 3.0, cotton, 100	Light dirt and half-load
SC02-002	9, 7.0, cotton, 100	Heavy dirt and full load
SC02-003	5, 2.0, delicate, 80	Delicate fabric must use safe spin
SC02-004	7, 5.0, synthetic, 25	Low-water situation
SC02-005	6, 8.5, heavy, 70	Over capacity; validation must reject

After entering the rows

- [] Open the CSV once in a text editor and once in a spreadsheet.
- [] Check that no heading contains an accidental space.
- [] Check that units are not mixed inside a numeric column.
- [] Check that categories use one spelling only.
- [] Check that identifiers are unique.
- [] Ask a member who did not create the file to explain three rows.

Write the comparison method

The baseline is deliberately simple. It lets you prove later whether the Soft Computing method is better.

1	Create one fixed preset for delicate, cotton, synthetic, and heavy fabric.
2	For each preset, write fixed wash_time_min, water_litre, detergent_ml, and spin_rpm.
3	Ignore dirt and water availability in the baseline. That limitation is intentional.
4	If load_kg is above machine capacity, do not return settings; show an error.

What to save

- [] Put the numbered rules in docs/STEP-1.md.
- [] Write the same rules as pseudocode in docs/baseline-pseudocode.md.
- [] For each of the five mandatory cases, write the expected baseline behavior.
- [] Mark every temporary threshold clearly.
- [] Do not call this baseline fuzzy logic, ANN, or GA.

Pseudocode format

```

READ one valid input row
IF the row is invalid:
  SHOW a clear error
ELSE:
  APPLY the simple rules in order
  CALCULATE the baseline output
  PRINT the output and the rule that was used
SAVE the result for later comparison

```

Step 1 requires the baseline design and expected behavior. A complete baseline program is built in Step 3, but writing a tiny runnable version now is encouraged.

Make the computer check your data

Install pandas and run the starter validator. The first version checks file shape, headings, missing values, and simple data types.

```
python -m venv .venv
# Windows PowerShell
.venv\Scripts\Activate.ps1
pip install pandas
pip freeze > requirements.txt
python src/validate_data.py
```

Starter content for src/validate_data.py

```
import pandas as pd

FILE = 'data/sample_input.csv'
REQUIRED = ['scenario_id', 'dirt_score', 'load_kg', 'fabric_type', 'water_availability_pct']
NUMERIC = ['dirt_score', 'load_kg', 'water_availability_pct']

df = pd.read_csv(FILE)
print('shape:', df.shape)
print('columns:', list(df.columns))
print('missing values:', df.isna().sum().to_dict())

missing_columns = [c for c in REQUIRED if c not in df.columns]
assert not missing_columns, f'Missing columns: {missing_columns}'
assert not df[REQUIRED].isna().any().any(), 'Missing value found'
for column in NUMERIC:
    assert pd.api.types.is_numeric_dtype(df[column]), f'{column} must be numeric'

print('STEP 1 DATA CHECK PASSED')
```

Add project-specific checks

- ☐ Every percentage must stay inside 0 to 100.
- ☐ Counts, time, distance, power, money, or rainfall cannot be negative.
- ☐ Every category must match the documented spelling.
- ☐ Every identifier must be unique.
- ☐ The known invalid/impossible case must fail with a clear message.
- ☐ The five valid mandatory cases must pass.

Take a screenshot of one PASS result and one expected FAIL result. Save them in results/step1/.

Draw the screen before coding it

Use paper, PowerPoint, Figma, Canva, or a simple drawing tool. The sketch is not the final UI.

Your screen must show

- [] Input area: Dirt score, load kg, fabric, water availability.
- [] One clear Run, Recommend, Predict, or Optimize button.
- [] Output area: Wash time, water, detergent, spin, baseline preset, warning.
- [] A place for validation errors.
- [] A place for one explanation or chart.
- [] A visible difference between baseline output and future Soft Computing output.

Simple wireframe to copy

```
+-----+
| PROJECT TITLE                |
+-----+
| INPUTS      | RESULTS      |
| [field] [value] | Baseline: ... |
| [field] [value] | Soft Computing: ... |
| [field] [value] | Explanation: ... |
| [ RUN / RECOMMEND ] | [ chart or visual area ] |
+-----+
| Validation / error message    |
+-----+
```

Save the sketch

1	Export or photograph the sketch clearly.
2	Save it as docs/product-v1-sketch.png.
3	Insert it in docs/STEP-1.md.
4	Ask one person outside the team to explain what they think the screen does.
5	Correct any label that the person does not understand.

Make small, honest commits

Each commit should contain one understandable improvement. Every member must make at least one commit.

Recommended commit sequence

1	Create repository structure for SC02
2	Add team roles and Step 1 project contract
3	Define input fields, units, and validation rules
4	Add validated sample wash scenarios
5	Add five mandatory test cases
6	Add sample-data validation script
7	Add Product V1 screen sketch
8	Add Step 1 evidence and run instructions

Before every push

- ☐ Pull the latest team version.
- ☐ Run the validation program.
- ☐ Check git status and add only intended files.
- ☐ Write a message that says what changed.
- ☐ Push your branch and ask another member to review.

Do not do this

- ☐ Do not use messages such as update, changes, final, or done.
- ☐ Do not upload the complete project in one last commit.
- ☐ Do not commit .venv, passwords, private datasets, or personal information.
- ☐ Do not overwrite another member's work without discussing it.

Exactly what to show faculty

Open the repository and demonstrate the following in order.

1	Show the repository URL and collaborator list.
2	Open README.md and state the one-sentence problem.
3	Open docs/STEP-1.md and explain the user, inputs, outputs, M1, and M2.
4	Open data/README.md and show the field meanings, units, source links, and assumptions.
5	Open the sample data and point out one normal, one boundary, and one difficult row.
6	Run <code>python src/validate_data.py</code> and show PASS.
7	Run or show the deliberately invalid case and explain the expected error.
8	Explain the baseline rules and expected result for one test case.
9	Show the Product V1 screen sketch.
10	Show Git history and one meaningful commit from every member.

Final Step 1 checklist

- ☐ Every required file exists and opens.
- ☐ The problem boundary is small and clear.
- ☐ Inputs and outputs have names, meanings, units, and rules.
- ☐ The starter data is understandable and valid.
- ☐ Five mandatory cases are written with expected behavior.
- ☐ The simple baseline is explainable.
- ☐ Validation shows both PASS and expected FAIL evidence.
- ☐ The screen sketch includes input, result, explanation, and errors.
- ☐ Every member has committed and can explain the work.

After faculty approval, begin Step 2: create the reproducible 10,000+ record/scenario pipeline. Do not change field meanings or units silently; update the contract and commit the change.

What comes after Step 1

Step	Stage	What happens
1	M1	Choose the washing machine capacity and define inputs.
2	M1	Create and validate realistic washing scenarios.
3	M1	Build fixed wash-cycle presets.
4	M1	Build fuzzy variables, rules, and outputs.
5	M1	Join the controller and screen as Product V1.
6	M2	Use GA to tune quality and resource trade-offs.
7	M2	Test delicate, scarce-water, and heavy-load cases.
8	M2	Deploy Product V2 and finish all evidence.

M1 finish line

- ☐ 10,000+ cited records or documented scenarios
- ☐ Working baseline
- ☐ Working core Soft Computing method
- ☐ Runnable Product V1
- ☐ At least two useful result visuals
- ☐ Interim report and M1 demonstration

M2 finish line

- ☐ Advanced or hybrid method
- ☐ Parameter experiments and difficult-case testing
- ☐ Baseline vs M1 vs M2 comparison
- ☐ Public deployment and usable interface
- ☐ Final report, approximately five-minute video, presentation, and viva preparation

A strong capstone is one that another person can open, run, understand, test, and compare.